15.4.5 Wrap-Up: Ticket Out the Door

1. There are 25 prime numbers between 1 and 100. There are 46 prime numbers between 1 and 200. Two events are shown.

Event A	Event B
A computer produces a random number between 1 and 100 that is prime.	A computer produces a random number between 1 and 200 that is prime.

Ava said Event B is more likely to occur because there are 46 possible outcomes in the event compared to only 25 in Event A. Oscar said Event A is more likely because $P(\text{A}) = 0.25$ and $P(\text{B}) = 0.23$.

a. Explain who is correct.

b. Write a short note to the student who was incorrect explaining the error they made.

Unit 15, Lesson 5: Experimental and Theoretical Probability

Benchmarks

MA.7.DP.2.3 Find the theoretical probability of an event related to a simple experiment.

MA.7.DP.2.4 Use a simulation of a simple experiment to find experimental probabilities and compare them to theoretical probabilities.

Learning Targets

- ☐ I can perform a simulation to determine an experimental probability.
- ☐ I can find the theoretical probability of an event.
- ☐ I can compare the experimental probability from a simulation to the theoretical probability of a simple experiment.

15.5.1 Warm-Up: Marbles

1. Fill in the blanks in the statements below so the probability of drawing a red marble from either bag is the same. Use any number 1 through 9 at most one time each.

   There are ______ red marbles and ______ blue marbles in Bag A. There are ______ red marbles and ______ green marbles in Bag B.

15.5.2 Exploration: Playing the Long Game

Samaria plays a game with a fair six-sided die in which she only wins if she rolls a 1 or a 2.

1. List the outcomes in the sample space for rolling the fair die.

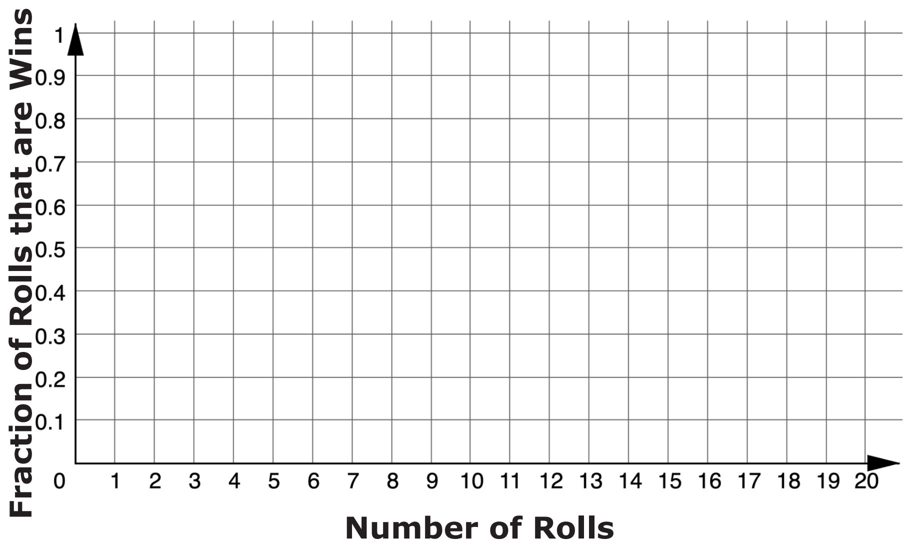
2. Discuss the probability of Samaria winning the game with your partner or group. Then, complete the statement.

   The probability Samaria wins the game, represented as a fraction, is _____ because there are _____ possible winning outcomes out of _____ total possible outcomes from rolling the die.

3. If Samaria is given the option to flip a coin and win if it comes up heads, explain whether or not that would be a better option for her to win using probability.

4. With your partner or group, follow these instructions 10 times to create the graph.
   - One person rolls the number cube. Everyone records the outcome.
   - Calculate the fraction of all rolls that are wins for Samaria so far. Approximate the fraction with a decimal value rounded to the hundredths place using a calculator. Record both the fraction and the decimal in the last column of the table.
   - On the graph, plot the number of rolls and the fraction of rolls that were wins.
   - Pass the number cube to the next person in the group.

Roll	Outcome	Total Number of Wins for Samaria	Fraction of Games Played that are Wins
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			



5. With your partner or group, discuss what you notice is happening with the points on the graph.

   a. After 10 rolls, what fraction of the total rolls were a win?

   b. How close is this fraction to the probability that Samaria will win?

6. Roll the number cube 10 more times. Record your results in this table and add them to the graph from earlier.

Roll	Outcome	Total Number of Wins for Samaria	Fraction of Games Played that are Wins
11			
12			
13			
14			
15			
16			
17			
18			
19			
20			

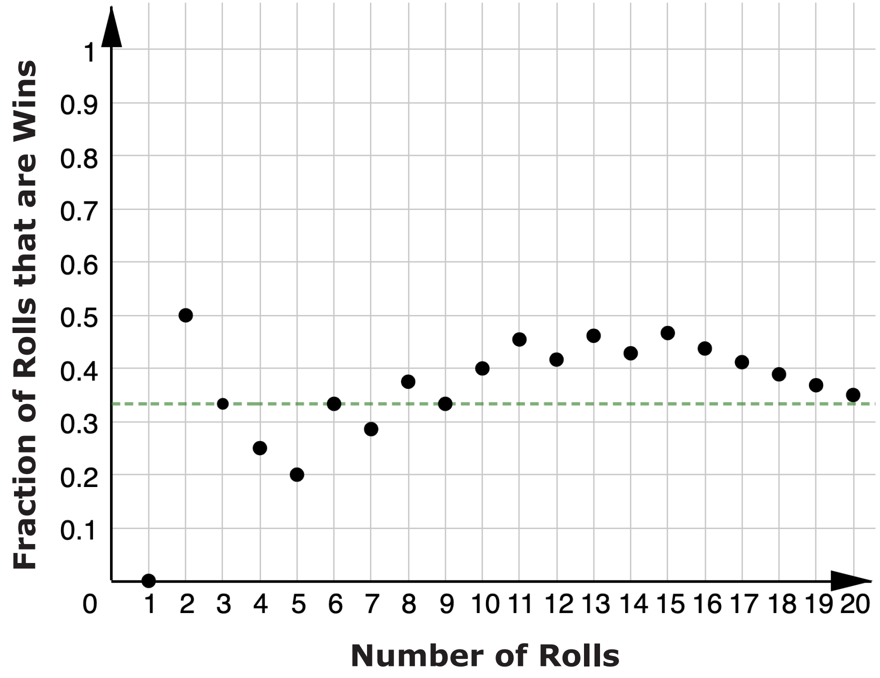
7. Observe any trends on the graph after 20 rolls.

    a. After 20 rolls, what fraction of the total rolls was a win?

    b. How close is this fraction to the probability that Samaria will win?

15.5.3 Bring It Together: Playing the Long Game

A sample graph from a group who performed the same simple experiment in the previous activity is shown.



1. What does the green dotted line represent on the graph?

2. Consider the data from all of the groups performing the experiment.

a. Complete the table below with your class by adding the overall data from each group. The first row has already been completed using the sample data above.

Group	Total Number of Wins for Samaria	Fraction of Games Played that are Wins
Sample Data	7	$\frac{7}{20} = 0.35$

b. Write the fraction that represents the total number of wins out of the total number of games played by all groups.

c. How close is this fraction to the probability that Samaria will win?

The probability for an event represents the proportion of the time it is expected for the event to occur in the long run, over many trials. It does not mean that is exactly how many times the event will actually occur.

15.5.4 Guided Instruction: Experimental and Theoretical Probability

Consider the terms defined below and how they relate to the previous Exploration.

A ***theoretical probability*** is a number between 0 and 1 that represents the likelihood of an event in a theoretical model based on a sample space. If all outcomes in the sample space are equally likely, then the theoretical probability of an event is the ratio of the number of outcomes in the event to the number of outcomes in the sample space.

An ***experimental probability*** is a number between 0 and 1 that shows the ratio of the number of times an event occurs to the total number of trials or times the activity is performed.

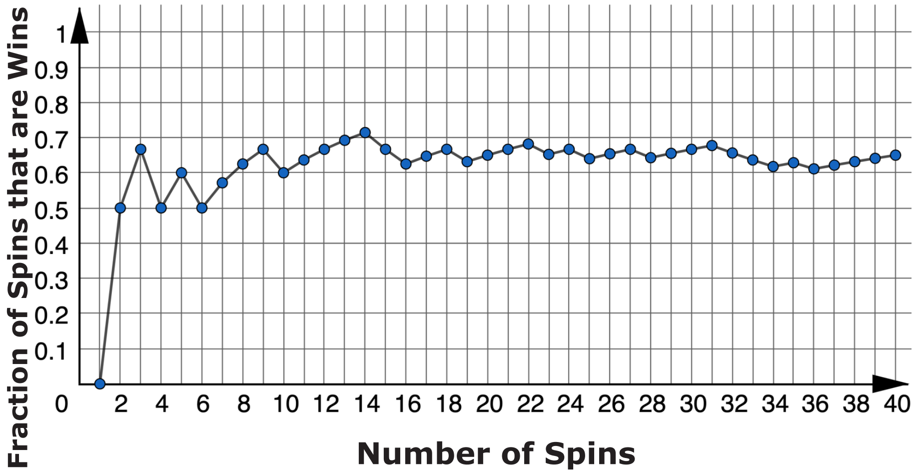
1. Use the definitions to complete the statements.

In the Exploration, the ○ experimental ○ theoretical probability of Samaria winning the game was $\frac{1}{3}$ because there were ○ 2 ○ 3 ○ 6 possible outcomes out of ○ 2 ○ 3 ○ 6 in the sample space that won. By perfoming the activity, each group determined ○ an experimental ○ a theoretical probability for the experiment.

15.5.5 Try It: Experimental and Theoretical Probability

1. Deepa wants to know if flipping a quarter really does have a probability of $\frac{1}{2}$ of landing heads-up, so she flips a quarter 10 times. It lands heads-up 3 times and tails-up 7 times. Explain whether or not she has proven that the probability of landing heads up is not $\frac{1}{2}$.

2. A spinner is spun 40 times for a game. The graph shows the fraction of games that are wins.



a. Estimate the probability of a spin winning this game based on the graph.

b. What type of probability is your estimate?

15.5.6 Wrap-Up: Cool Down

1. Your friend was absent today and will need the notes from class. Write a quick note to help them understand the difference between the theoretical and experimental probability of a simple experiment.

Unit 15, Lesson 6: Simulating Small and Large Numbers of Trials

Benchmark

MA.7.DP.2.4 Use a simulation of a simple experiment to find experimental probabilities and compare them to theoretical probabilities.

Learning Targets

- ☐ I can compare the experimental probability from a simulation to the theoretical probability of a simple experiment.
- ☐ I can use results from computer-simulated events to informally define the Law of Large Numbers.